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Profile

- 📌 I am a theoretical physicist working as a post-doctoral fellow at Institute of Physics Bhubaneswar, India. My main research focus is on the study of strongly interacting matter under extreme conditions, with a special emphasis on its relevance to neutron/hybrid stars (NS/HS)s. More specific the current and the near future research interest is to study the strongly interacting matter in the context of NSs, merging of NSs etc. constraints with gravitational wave (GW)s and NICER observations. Currently, I have **13 research papers** on arXiv, **10 out of them** are published in different international journals. I have developed all the codes used in my research by myself from the scratch to get the important results.

Education

- 2017 – 2022 📌 **Ph.D., Indian Institute of Technology Gandhinagar, India** (Physical Research Laboratory Ahmedabad, India).
Thesis title: *Properties of strongly interacting matter under extreme conditions*
Defense date: 15 May 2023
Supervisor: Prof. Hiranmaya Mishra (hiranmaya@niser.ac.in)
- 2014 – 2016 📌 **M.Sc. Banaras Hindu University, Varanasi, India** in Physics
Thesis title: *The field configuration in the Gaseous Electron Multiplier (GEM) Detector.*
- 2011 – 2014 📌 **B.Sc. Banaras Hindu University, Varanasi, India** in Physics

Employment History

- Oct 24 – Till date 📌 **Post-doctoral fellow** at Institute of Physics Bhubaneswar, India
- Jun 23 – Aug 24 📌 **Post-doctoral fellow** at India Institute of Science Education and Research Bhopal, India
- Sep 22 – May 23 📌 **Post-doctoral fellow** at Physical Research Laboratory Ahmedabad, India

Skills

- Languages 📌 Strong reading, writing and speaking competencies for English, Hindi and Sanskrit
- Coding 📌 Python, FORTRAN, $\text{\LaTeX}$, C, C++, ROOT, Linux Shell scripting, Html, css
- Software 📌 Mathematica, Maxwell2D, ANSYS3D
- Misc. 📌 Teaching, Playing Volleyball and Table-Tennis

Research Projects

Cosmos Sounds: Unveiling Dark Matter's Influence on Neutron Star Characteristics via Gravitational Wave Signatures DOI: <https://doi.org/10.54499/2023.14108.CPCA.A3>.

Funding Program: FCT_CPCA_2023_01

Knowledge Fields: Exact sciences/Physical sciences

Main Institutions and Funder Country: Universidade de Coimbra (UC), Portugal

References

Prof. Hiranmaya Mishra

Professor

School of Physical Sciences, NISER, Bhubaneswar,
P.O. Jatani, Khurda, Odisha, India - 752 050.

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Prof. Constança Providência

Professor

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Dr. Ritam Mallick

Associate Professor

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Education and Research Bhopal,
Bhauri, Madhya Pradesh, India - 462066.

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List of Publications

Journal Articles (Published and/or accepted)

- 1 **D. Kumar**, A. Karan, A. Verma, H. Mishra, and R. Mallick, *Modification of the universal relation between mass, radius, and nonradial f -mode oscillation in proto-neutron stars*, 2025. DOI: [10.1103/PhysRevC.111.055805](https://doi.org/10.1103/PhysRevC.111.055805). arXiv: [2409.01785](https://arxiv.org/abs/2409.01785) [astro-ph.HE].
- 2 **D. Kumar**, T. Malik, and H. Mishra, *The footprint of nuclear saturation properties on the neutron star f mode oscillation frequencies: a machine learning approach*, 2025. DOI: [10.1088/1475-7516/2025/03/033](https://doi.org/10.1088/1475-7516/2025/03/033). arXiv: [2402.03054](https://arxiv.org/abs/2402.03054) [nucl-th].
- 3 **D. Kumar**, T. Malik, H. Mishra, and C. Providência, *Constraints on maximum neutron star mass from protoneutron star evolution*, 2025. DOI: [10.1103/zy4h-6wj2](https://doi.org/10.1103/zy4h-6wj2). arXiv: [2505.18888](https://arxiv.org/abs/2505.18888) [nucl-th].
- 4 S. Tewari, S. Chatterjee, **D. Kumar**, and R. Mallick, *Analyzing the dense matter equation of states in the light of the compact object HESS J1731-347*, 2025. DOI: [10.1103/PhysRevD.111.103009](https://doi.org/10.1103/PhysRevD.111.103009). arXiv: [2410.20355](https://arxiv.org/abs/2410.20355) [astro-ph.HE].
- 5 **D. Kumar**, T. Malik, H. Mishra, and C. Providencia, *Robust universal relations in neutron star asteroseismology*, 2023. DOI: [10.1103/PhysRevD.108.083008](https://doi.org/10.1103/PhysRevD.108.083008). arXiv: [2306.09277](https://arxiv.org/abs/2306.09277) [nucl-th].
- 6 **D. Kumar**, H. Mishra, and T. Malik, *Non-radial oscillation modes in hybrid stars: consequences of a mixed phase*, 2023. DOI: [10.1088/1475-7516/2023/02/015](https://doi.org/10.1088/1475-7516/2023/02/015). arXiv: [2110.00324](https://arxiv.org/abs/2110.00324) [hep-ph].
- 7 A. Abhishek, A. Das, **D. Kumar**, and H. Mishra, *Thermoelectric transport coefficients of quark matter*, 2022. DOI: [10.1140/epjc/s10052-022-09999-z](https://doi.org/10.1140/epjc/s10052-022-09999-z). arXiv: [2007.14757](https://arxiv.org/abs/2007.14757) [hep-ph].
- 8 A. Das, **D. Kumar**, and H. Mishra, *Chiral susceptibility in the Nambu–Jona-Lasinio model: A Wigner function approach*, 2019. DOI: [10.1103/PhysRevD.100.094030](https://doi.org/10.1103/PhysRevD.100.094030). arXiv: [1907.12332](https://arxiv.org/abs/1907.12332) [hep-ph].

Journal Articles (Communicated)

- 1 **D. Kumar** and P. K. Sahu, *Constraining the neutron star equation of state by including the isoscalar-vector and isovector-vector coupling using the Bayesian analysis*, May 2025. arXiv: [2505.02618](https://arxiv.org/abs/2505.02618) [nucl-th].
- 2 **D. Kumar** and H. Mishra, *CP violation in cold dense quark matter and axion effects on the non-radial oscillations of neutron stars*, Nov. 2024. arXiv: [2411.17828](https://arxiv.org/abs/2411.17828) [hep-ph].

Conference Proceedings

- 1 **D. Kumar** and H. Mishra, “Axion effects on the non-radial oscillations of neutron stars.” Submitted: (XXVIDAE–BRNSHighEnergyPhysicsSymposium2024).

- 2 **D. Kumar** and H. Mishra, "Axion effects on the non-radial oscillations of neutron stars." Submitted: ([JournalOfSubatomicParticlesandCosmology:ATHIC2025](#)).
- 3 **D. Kumar** and P. K. Sahu, "Constraining the neutron star equation of state by including the isoscalar-vector and isovector-vector coupling using the Bayesian approach." Submitted: ([XXVIDAE-BRNSHighEnergyPhysicsSymposium2024](#)).
- 4 **D. Kumar**, H. Mishra, and T. Malik, "Quark Matter in the Core of Neutron Stars and Its Consequences on Non-radial Oscillations," vol. 304, 2024, pp. 637–639.  DOI: [10.1007/978-981-97-0289-3_144](https://doi.org/10.1007/978-981-97-0289-3_144).
- 5 A. Jaiswal *et al.*, "Dynamics of QCD matter — current status," o2, vol. 30, 2021, p. 2 130 001.  DOI: [10.1142/S0218301321300010](https://doi.org/10.1142/S0218301321300010). arXiv: [2007.14959](https://arxiv.org/abs/2007.14959) [[hep-ph](#)].

Journal Articles (Under Preparation)

- 1 **D. Kumar** and H. Mishra, *Nambu–Jona-Lasinio model parameters from the astrophysical observations using Bayesian analysis*.
- 2 **D. Kumar** and P. K. Sahu, *Study of supernovae*.
- 3 **D. Kumar**, A. Verma, H. Mishra, and R. Mallick, *Shock waves in the proto-neutron stars*.
- 4 A. Das, **D. Kumar**, T. Malik, and H. Mishra, *The asteroseismology in neutron stars in presence of dark matter*.
- 5 A. Karan, **D. Kumar**, and R. Mallick, *Non-radial oscillations in two fluid approach with dark matter in the core of neutron star*.

Miscellaneous Experience

Awards and Achievements

JAM 2014	 Joint Admission Test for M.Sc.
JEST 2017	 Joint Entrance Screening Test
GATE 2019	 Graduate Aptitude Test in Engineering
JRF 2017	 Junior Research Fellow at PRL, Ahmedabad
SRF 2019	 Senior Research Fellow at PRL, Ahmedabad
PDF 2023	 PDF at PRL, Ahmedabad from Sept 2022 to May 2023 with Prof. Srubabati Goswami
PDF 2024	 PDF at IISER Bhopal from June 2023 to Aug 2024 with Dr. Ritam Mallick

Current Employment

-  **Institute Post-doctoral Fellow from Oct 2024 - till date.** Nuclear Physics Group, Institute of Physics Bhubaneswar, India - 751005

Science Activity

-  I was one of the members of the organizing committee of the student conferences in my institute and designed the meeting webpage. ([YPM 2019](#), [YPM 2020](#)).
-  I have actively participated in the science exhibition programs organized by my institute where I have to explain my research work to the common people.

Miscellaneous Experience (continued)

Scientific Visits (National/International)

- I visited **Prof. Hiranmaya Mishra (visiting Prof)** at **School of Physical Sciences, National Institute of Science Education and Research, Bhubaneswar, India, 752 050** in (Sep 2023, Jul 2023, Oct 2023, Sep 2024) and completed arXiv:2402.03054, arXiv:2411.17828 works.
- I also visited **Prof. Constanca Providencia and Dr. Tuhin Malik** at **Departamento de Fisica, Universidade de Coimbra, 3004-516 Coimbra, Portugal** in Jun 2024 and completed one work on evolution proto-neutron star (current status: writing).
- Along with Coimbra visit, I visited **Prof. Kostas K Kokkotas, Department of Physics, EBERHARD KARLS University of Tuebingen, Germany**. There I started working on the effects of dark matter on the non-radial oscillation mode of exotic neutron stars. The work is still going on.
- Recently, I have visited **Prof. Gargi Chaudhuri, Variable energy cyclotron centre, BARC Institute, Kolkata, India** in Oct 2024 and started collaboration.

Workshops/Conferences

National/International

- **The Myriad Colorful Ways of Understanding Extreme QCD Matter**
ICTS, Bangalore - August 2019
- **Workshop on High Energy Physics Phenomenology (WHEPP) XVI**
IITG, Guwahati - December 2019
- **Virtual Meeting on Compact Stars and QCD 2020**
(Online, ICTS) 17-21 August 2020
- **International Conference on Critical Point and Onset of Deconfinement (CPOD2021)**
(Online) 15-19 March 2021
- **Neutron stars as multi-messenger laboratories for dense matter**
(Online, Virtual/Online) 14-17 June 2021
- **Quantum Information in QFT and AdS/CFT-II**
(Online, IITGN) 18-20 August 2021
- **Neutron stars as multi-messenger laboratories for dense matter**
(Online, Villa Tambosi, Trento & Zoom) 20-24 June, 2022
- **The 9th International Conference on Quarks and Nuclear Physics**
(Online, Florida State University & Zoom) 05-09 Sept, 2022
- **XXV DAE-BRNS HEP SYMPOSIUM**
(IISER MOHALI) 12-16 Dec, 2022
- **XIth International Symposium on Nuclear Symmetry Energy**
(GSI online) 18-22 September 2023
- **2nd workshop on DYNAMICS OF QCD MATTER**
(School of physical science, NISER, Bhubaneshwar) 07-09 October 2023
- **Aspects of QCD phase diagram**
(IISER, Bhopal) 18-20 November 2023

Workshops/Conferences (continued)

- **Phase transitions - From the Laboratories to the Cosmos**
60th Karpacz winter school on theoretical physics and WE-Heraeus Physical school
(Institute of Theoretical Physics, University of Wroclaw / Hotel "Artus", Karpacz) 16-25 May 2024
- **Karpacz winter Kindergarten on theoretical physics**
(Institute of Theoretical Physics, University of Wroclaw / Holiday and Conference Centre "Raven Rocks" ("Krucze Skaly, Karpacz) 19-25 May 2024
- **XXVI DAE-BRNS HEP SYMPOSIUM**
(Banaras Hindu University, Varanasi, India) 19-23 Dec, 2024.
- **10th Asian Triangle Heavy-Ion Conference - ATHIC 2025**
(Indian Institute of Science Education and Research Bhehrampur, India) Jan, 2025.
- **New Frontiers in Extreme Nuclear Matter: Understanding the Intersection of Nuclear and Particle Physics**
(Institute of Physics Bhubaneswar, India) Feb, 2025.
- **Hot QCD Matter 2025 (Series 3)**
(Indian Institute of Technology Bhilai, India) Sep, 2025.
- **India-JINR Workshop on Particle, Nuclear, Neutrino Physics and Astrophysics**
(National Institute of Science Education and Research Bhubaneswar, India) 10-12 Nov, 2025.

Presentations (Talks)

- **Chiral Susceptibility in NJL model: A Wigner Function Approach**
Young Physics Meet (YPM) 2019, PRL, Ahmedabad
- **Chiral Susceptibility in Quark Matter**
WHEPP XVI 2019, IIT Guwahati
- **Quarkyonic Matter in the core of Neutron Star**
- **Non-Radial Oscillation Modes in Hybrid Star: Consequences of Mixed Phase**
- **Strongly interacting matter under extreme conditions**
- **Thermoelectric transport coefficients of quark matter**
- **Correlations between nuclear matter quantities and different attributes of neutron star**
- **The footprint of nuclear saturation properties on the neutron star f mode oscillation frequencies: a machine learning approach**
(60th Karpacz winter school on theoretical physics)
- **Study of correlations of nuclear saturation properties and neutron star f mode oscillations from a machine learning**
(Karpacz winter Kindergarten on theoretical physics)
- **Correlations between nuclear matter saturation properties and non-radial f mode oscillations of neutron stars**
(EBERHARD KARLS University of Tuebingen, Germany)
- **Correlations between nuclear matter saturation properties and non-radial f mode oscillations of neutron stars**
(Institute of Physics Bhubaneswar, India, 18-09-2024)
- **Constraining the neutron star equation of state by including the isoscalar-vector and isovector-vector coupling using the Bayesian approach**
(XXVI DAE-BRNS HEP SYMPOSIUM, Banaras Hindu University, Varanasi, India, 22-12-2024)

Workshops/Conferences (continued)

- **Axion effects on the non-radial oscillations of neutron stars**
(XXVI DAE-BRNS HEP SYMPOSIUM, Banaras Hindu University, Varanasi, India, 22-12-2024)
- **Axion effects on the non-radial oscillations of neutron stars**
(10th Asian Triangle Heavy-Ion Conference - ATHIC 2025)
- **Neutron star equation of state from the Bayesian analysis**
(Hot QCD Matter 2025 (Series 3))
- **Neutron star matter equation of state from the Bayesian analysis**
(India-JINR Workshop on Particle, Nuclear, Neutrino Physics and Astrophysics)
- **Constraining the nuclear matter equation of state in the light of gravitational wave observation**
(New Frontiers in Extreme Nuclear Matter: Understanding the Intersection of Nuclear and Particle Physics)

Presentations (Posters)

- **Quark matter in the core of neutron stars and its consequences on non-radial oscillations**
DAE-HEP 2022 - XXV DAE-BRNS HEP SYMPOSIUM (IISER MOHALI) 12-16 Dec, 2022
- **Robust universal relations in neutron star asteroseismology**
NuSym23, XIth International Symposium on Nuclear Symmetry Energy (18-22 September 2023)

Description of research interests and achievements

Thesis in Brief

- In my thesis, I worked on different regions of the QCD phase diagram (in $T - \mu$ plane) - (i) non-vanishing temperature and vanishing baryon chemical potential, (ii) non-vanishing temperature and non-vanishing baryon chemical potential, and (iii) vanishing temperature and non-vanishing baryon chemical potential regions. In the first region, we have studied chiral phase transition of quark matter (QM) at extremely high temperature at vanishing baryon chemical potential but non-zero chiral chemical potential. We also worked on some transport properties like electrical conductivity, thermal conductivity and Seebeck effects of QM at non-zero temperature, non-zero chemical potential in the presence of external non-zero magnetic field in the second region. In the third region, we worked on the hadron-quark phase transition at zero temperature and extremely high density (few times nuclear saturation density, $\rho_0 = 0.16\text{fm}^{-3}$) in the context of neutron/hybrid stars. We also studied non-radial oscillation modes (g and f modes) in neutron/hybrid stars to understand the type of matter inside the star. We find some robust universal relations, relating mass, radius and non-radial oscillation (f) mode frequency of NSs using about 30 million EOSs (DDB and DDB-Hyb). The DDB set is the set of relativistic mean field (RMF) equations of states (EOSs) where the couplings are density dependent and estimated through Bayesian analysis while the DDB-Hyb set is the set where low density part of EOS is the DDB EOS and high density (asymptotically high density) pQCD EOS, the intermediate region is interpolated by piecewise polytrope. We also find a new universal relation for fixed mass neutron/hybrid star relating non-radial oscillation f -mode frequency and radius of the neutron/hybrid star.

Description of research interests and achievements (continued)

Current and Future Research Interests and Experiences

■ My research focuses on the physics of dense matter under extreme conditions, with particular emphasis on the equation of state (EoS) of neutron star matter and its manifestations in both astrophysical and heavy-ion environments. I integrate theoretical nuclear physics, QCD-inspired models, and modern computational techniques to address fundamental questions about matter at supra-nuclear densities, while also bridging connections to gravitational-wave observations and laboratory experiments. During my doctoral research, I investigated the EoS of neutron star matter by combining hadronic and quark-level descriptions. My work established the following contributions:

- **Relativistic Mean Field (RMF) Models for Nuclear Matter:** Developed and constrained models that reproduce saturation properties of nuclear matter and extended them to astrophysical densities relevant for neutron stars.
- **Nambu–Jona-Lasinio (NJL) Models for Quark Matter:** Explored quark matter properties under conditions of high baryon density and investigated the role of chiral symmetry breaking.
- **Hadron–Quark Phase Transition:** Applied Gibbs construction mechanisms to model the transition from hadronic to quark phases, identifying signals of hybrid star configurations and twin-star solutions. Studied similarities between matter in relativistic heavy-ion collisions and the dense environments in neutron stars.
- **Axion-Like Particles in Neutron Stars:** Examined the role of beyond-Standard-Model physics, particularly axion-like particles, in modifying neutron star oscillation properties.

In addition, I have contributed to studies of transport phenomena (thermal and electrical conductivity, Seaback coefficient in hot and dense matter), connecting microscopic dynamics with macroscopic observables. These works have resulted in peer-reviewed publications and presentations at international conferences. My present research extends these efforts by incorporating Bayesian inference, machine learning, and artificial intelligence to systematically constrain the neutron star EoS using multi-messenger observations. By combining nuclear theory with data from gravitational waves, pulsar timing, and X-ray measurements, I aim to narrow down the uncertainties in dense matter physics. As a postdoctoral researcher, I plan to pursue the following directions:

- **Binary Neutron Star Mergers:** Investigate the role of hadron–quark transitions in the post-merger dynamics, gravitational-wave signatures, and possible electromagnetic counterparts. This includes modeling the interplay of microphysics with macroscopic merger dynamics.
- **Non-Radial Oscillations of Neutron Stars:** Develop a framework for studying oscillation spectra as probes of the EoS, transport properties, and exotic matter phases. These oscillations can serve as a diagnostic tool to distinguish between purely hadronic stars, hybrid stars, and stars with axion-modified cooling.
- **AI and Bayesian Frameworks:** Advance the use of machine learning and Bayesian techniques to connect theoretical models of dense matter with multi-messenger constraints, enabling predictive and interpretable frameworks for neutron star physics.
- **Interdisciplinary Connections:** Explore connections with heavy-ion collisions, where transport coefficients and chiral dynamics can be constrained in complementary regimes.

Description of research interests and achievements (continued)

Experiences toward Research

- I explored the high temperature, where we find the hadron-quark cross-over in the context of heavy-ion collisions, and zero temperature where we find hadron-quark phase transition in the context of compact stars. I develop expertise of finding the equations of state for the matter present in the core of compact stars. I developed nuclear matter equation of state for the cold and proto-neutron stars in the relativistic mean field model with and without non-linear terms in the Lagrangian. In case of without non-linear terms we have taken the density dependent coupling while in the other the couplings are constant over the density. I also develop the quark matter equations of states in the Nambu–Jona-Lasinio (NJL) and MIT bag models where the NJL model defines chiral phase transition and fails to define the color confinement while MIT bag model defines the color confinement and fails to define the chiral phase transition. I developed the expertise to study the stars (radial and non-radial) oscillations of cold and proto-neutron stars. Currently, I have developed the equation of state of dark matter where dark matter interacts with Higgs field and Higgs field interacts with normal matter. So the dark matter interacts with normal matter indirectly. I am very good in some computer languages for example Fortran, python and c++, etc. I have also worked in a Mathematica to solve some definite integrals. Currently, I am writing some codes to find the oscillations of neutron stars in two fluids scenarios.

Broader Impact and Vision

- My long-term goal is to establish a comprehensive research program that integrates nuclear theory, astrophysics, and computational methods to address open questions about dense QCD matter. By combining effective field theories, quark-level models, and AI-driven approaches, I aim to uncover the microphysics governing neutron stars and to provide testable predictions for upcoming gravitational-wave and electromagnetic observations. Equally, I am committed to fostering collaborative and interdisciplinary science. My research naturally interfaces with nuclear theory, astrophysics, gravitational physics, and particle phenomenology, providing opportunities to build bridges between communities. I also value mentoring and teaching, having guided students in computational and theoretical methods, and I see training the next generation of researchers as integral to my scientific vision. In summary, my expertise in modeling dense matter with RMF and NJL-type models, my work on hadron–quark transitions and axion physics, and my current use of AI and Bayesian frameworks position me strongly to contribute to ongoing efforts in neutron star astrophysics. A postdoctoral position in your group would provide the ideal platform to expand this research, collaborate across disciplines, and explore the fundamental physics of matter under extreme conditions.

A short summary of recent research achievements

Recent accomplishments

- *I have studied the **relativistic mean field (RMF)** theory of nuclear/hadronic matter at zero and non-zero temperature and baryon density. Along with that, I have also studied the **Nambu–Jona-Lasinio** model and **MIT bag** model to study the quark matter at zero and non-zero temperature and baryon density. For the hadron to quark phase transition, I have used **Gibbs** construction mechanism. All of the codes are written from the scratch in both **python** and **FORTRAN**. I have used various python libraries like **numpy**, **pandas**, **scipy** etc. These theories and models, I along with other collaborators studied the microscopic and macroscopic properties of **neutron/hybrid stars** and the various **non-radial oscillation** modes. For more details, see [14], [5], [6],*

A short summary of recent research achievements (continued)

- *In the context of **heavy-ion collisions**, I have studied the **chiral phase transition** at extreme temperature and zero baryon chemical potential but non-zero chiral chemical potential. We evaluated the expression of the Wigner function, which is a quantum analogue of classical phase-space distribution function. We used it to get the chiral condensate, which works as an order parameter to distinguish two phases and also define the effective mass of quark within the **Nambu–Jona-Lasinio** model. Along with the **chiral phase transition (crossover)**, we have studied the **various transport properties** for example **electrical conductivity, thermal conductivity, Seeback coefficient** etc of quark matter which is under extreme temperature non-zero chemical potential and magnetic field. For more details, see [7], [15], [8],*
- *I have studied the nuclear matter **equations of states** which satisfy the **nuclear matter saturation properties** like incompressibility of symmetric matter, symmetry energy, slope parameter, and incompressibility of symmetry energy at saturation density. Along with these saturation properties, the equations of states also satisfy the **astrophysical observations** of compact stars. These equations of state are described in the relativistic mean field (RMF) theory of nuclear matter and their coupling parameters are found using the **Bayesian approach**. After getting such equations of states for nuclear matter, we have also applied the **perturbative QCD** constraints on fitted the intermediate densities using the **polytropes**. For the full set of equations of state, we have determined various **universal relations** among mass, radius, and the f-mode non-radial oscillation frequencies. For further details, see [2], [4], [5],*
- *I, along with the collaborators, have applied the **machine learning** approaches to determine what kind of saturation property can be **easily probe** in the **neutron stars observations**. We have used the very famous **principal component analysis** and **random forest** method to determine the effects of various nuclear matter saturation properties on non-radial oscillations of neutron stars. For further details, see [2],*
- *I have estimated the temperature dependent equation of state to study the isothermal and isoentropic neutron stars. I, along with the collaborators, have found the temperature dependent universal relations in between radius, mass and f-mode frequency of isothermal neutron stars. For more details, see [1],*

On going research works

- *Currently, there are few projects on which I am working on and they are listed in the journal articles which are still under preparation in the section research publications. In brief, I am working on the asteroseismology of neutron stars in the presence of exotic matter (dark matter) and asteroseismology of proto-neutron stars. We are also working on the non-radial oscillations in two fluid scenario. The most general equation of state of nuclear matter at few times nuclear saturation density where hadron-quark phase transition happens. Using the observations, one could constrain the quark matter model too by using the Bayesian analysis. There is also a possibility of having the exotic type of matter in the core of neutron/hybrid/exotic stars. The study of shockwave inside the neutron stars could also give the information of the particle pattern at the surface on neutron stars. [16]–[20],*